Q18

Find the continued product by using the relevent formulas:

i $(2x^2+3y^2)(4x^4-6x^2y^2+9y^4)$

we will Use here formula:

$a^3+b^3 = (a+b)(a^2-ab+b^2)$

$(\underbrace{2x^2}_{a}+\underbrace{3y^2}_{b})\ (2x^2)^2-(2x^2)(3y^2)+(3y^2)^2$

$(2x^2+3y^2)(4x^4-6x^2y^2+9y^4)$

so there for Now we can also write

$(2x^2+3y^2)(4x^4-6x^2y^2+9y^4) = (2x^2)^3+(3y^2)^3$

$(2x^2+3y^2)(4x^4-6x^2y^2+9y^4) = 8x^6+2y^6$

Ans.